

Setting Up a Gateway Network Certificate

Users can add their own certificates for Incoming GAN connections by creating a metro-keystore using their own CA. The following steps describe this process, with the assumption that a Java JDK and OpenSSL are installed.

1. Navigate to a Java installation in a command window. You can use the Gateway's runtime JRE.

```
Java
cd "C:\Program Files\Inductive Automation\Ignition\lib\runtime\jre-win\bin"
```

Note: The Java installation file path may be different depending on the OS used. Refer to the path below for what fields will be changed:

```
Java
cd "<Ignition Home Folder>/lib/runtime/jre-<OS Type>/bin"
```

Options for <OS Type> could be one of the following:

- aarch64
- mac
- nix (sic)
- win

2. Use the keytool program to create a new keystore. Note that the user cannot use the 5 character default `metro` password since it is too short. Recommend users create more complex, secure passwords with this step.

```
Java
keytool -genkey -alias metro-key -keyalg RSA -keysize 4096 -keystore
"C:\tmp\gan-cert\new-metro-keystore"
```

Note: This code snippet above uses a temporary folder location for simplicity. Users should be encouraged to create a new permanent location for this information. Creating a tmp location isn't standardized for all OSs and there is also a chance that it may be removed on a restart.

Also, the keysize in this example was increased from the default key size of 2048.

3. Generate the certificate signing request, using the password created in step 2. Depending on the certificate the user is requesting, they may need to include the Subject Alternative Name (SAN) information within the request. This will include the DNS Name, the IP Address, and the URI.

```
Java
keytool -certreq -alias metro-key -file "C:\tmp\gan-cert\metro.csr" -keystore
"C:\tmp\gan-cert\new-metro-keystore" -ext
SAN=dns:test.internal,ip:127.0.0.1,uri:test.internal/metro
```

- If a user has multiple DNS and IPs they want to add to the certificate, say for a Master and Backup Gateway, they can be included under the same SAN:

```
Java
-ext
SAN=dns:test.internal,dns:localhost,ip:127.0.0.0,ip:127.0.0.1,uri:test.internal/me
tro
```

Note: SAN information is not always a requirement and sometimes won't even be accepted by a certificate authority. This [forum post](#) has a good explanation of what a SAN is and what details it shares.

Users may need to show proof of ownership of the domain before a certificate can be created. Learn more about how to do this in the [Ignition User Manual](#).

4. Send the **metro.csr** file to a CA authority. After processing, the authority will return signed certificates.

5. Backup and rename the original GAN keystore, and use the new keystore in its place.

```
Java
cd "C:\Program Files\Inductive Automation\Ignition\webserver"
mv .\metro-keystore metro-keystore.orig
cp "C:\tmp\gan-cert\new-metro-keystore" "C:\Program Files\Inductive
Automation\Ignition\webserver\metro-keystore"
```

6. Import the certificates provided by the CA authority into Ignition's JRE runtime. Make sure to follow the order below exactly.

Java

```
keytool -import -trustcacerts -alias root -file "C:\tmp\gan-cert\root-ca.crt"  
-keystore "C:\Program Files\Inductive  
Automation\Ignition\webserver\metro-keystore"
```

```
keytool -import -trustcacerts -alias INTER -file  
"C:\tmp\gan-cert\intermediate.crt" -keystore "C:\Program Files\Inductive  
Automation\Ignition\webserver\metro-keystore"
```

```
keytool -import -trustcacerts -alias metro-key -file  
"C:\tmp\gan-cert\new-metro-cert.crt" -keystore "C:\Program Files\Inductive  
Automation\Ignition\webserver\metro-keystore"
```

7. Modify the **data/ignition.conf** file to use the new metro keystore password.

Java

```
wrapper.java.additional.7=-Dmetro.keystore.password=<newPassword>
```

8. If using Ignition 8.1.13 or earlier, be sure to rename **data/certificates/cert-chain-store** to **cert-chain-store.orig** since the keystore password changed. This step can be skipped for 8.1.14+ as the file no longer exists. The file location described [here](#) will automatically trust the certificate.

9. Restart the Gateway and navigate to **Config > Gateway Network** and click the Diagnostics tab to confirm the local certificate information.